

Kria Yocto Vitis Vision Library App

Creating Vitis Platform and System Project

A. Overview

The Vitis Acceleration flow on Kria will be as of three steps as mentioned here - https://github.com/Xilinx/Vitis-Tutorials/blob/2025.1/Vitis_Platform_Creation/Design_Tutorials/01-Edge-KV260/README.md

B. Creating VIVADO Design

As this kria app follows the Vitis Acceleration flow, we will have to first create VIVADO Platform, use that platform to make the Vitis Extensible Platform.

With the Vitis Extensible Platform, we will be running compiled HLS kernel of ColorDetect (.xo kernel) and create the system project.

At the end of system project, we will be getting XCLBIN file.

We will copy XCLBIN, device tree file and shell.json as a firmware file. Also copy the compiled host application and then we can run it on KV260 Raspberry Pi v2 setup.

1. VIVADO Block design
2. VIVADO Address Editor
3. Constraint
4. Now create HDL wrapper , let Vivado do the step.
5. Generate the bitstream.

6. Export hardware, including bitstream. Here you will get the XSA file.

C. Creating Vitis Platform

D. Compile HLS Kernel

E. Creating and Build the System Project

Revision History

S.N	Revision Number	Revision Details	Author
1.	v1.0	Initial Revision	Krishna Gaihre and Sanam Shakya